Pediatric Pulseless Arrest

PC-04 | Pediatric Cardiac

Quick flow

01 Pediatric Arrest [EMT / AEMT / Paramedic]

Confirm pulselessness, age pathway, DNR/MOST, and obvious-death criteria. Use AO-02 through 3 days of age; use adult cardiac arrest at age 16 years or older.

02 High-Quality CPR + BVM [EMT / AEMT / Paramedic]

100-120/min • depth at least ⅓ chest AP diameter • full recoil • pauses under 10 sec. Use 30:2 with one rescuer or 15:2 with two rescuers.

03 AED / Monitor [EMT / AEMT / Paramedic]

Apply immediately. Use pediatric pads/attenuator when available, but do not delay defibrillation if standard equipment is immediately available.

04 Rhythm-Specific Pathway [EMT / AEMT / Paramedic]

VF/pulseless VT: PC-07. Asystole/PEA: PC-01. Obtain IV/IO promptly without delaying CPR or ventilation.

05 Suspected Opioid Cause? [AEMT / Paramedic]

Give naloxone while continuing uninterrupted CPR, oxygenation, and ventilation. Naloxone is not a substitute for resuscitation.

06 ROSC? [EMT / AEMT / Paramedic]

Move immediately to PC-08 Pediatric Post-Resuscitation Care; maintain airway support and notify the pediatric-capable destination early.

Overview

- Immediate routing protocol for a pulseless patient age 3 days through 15 years. Use AO-02 Newly Born through 3 days of age and the adult cardiac-arrest pathway at age 16 years or older.
- High-quality CPR and effective ventilation are central interventions because most pediatric arrests are respiratory/asphyxial in origin.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Begin high-quality CPR with effective BVM ventilation, apply AED/monitor, and use the applicable shockable or nonshockable rhythm pathway.
- Change compressors about every 2 minutes, minimize all interruptions, and do not delay CPR, ventilation, or defibrillation for advanced procedures.
- Use a pediatric length- or weight-based resuscitation system and obtain caregiver history when it does not interfere with resuscitation.

AEMT

- Perform all EMT care plus rapid IV/IO access, glucose and temperature assessment, authorized naloxone, and advanced airway support within current standing orders.
- Use IO promptly if IV access is not quickly available. Do not repeat IV attempts while delaying CPR or ventilation.

Paramedic

- Perform all prior care plus rhythm-specific manual treatment, continuous waveform $EtCO_2$ after an advanced airway, reversible-cause management, and post-ROSC transition.
- Do not interrupt compressions to intubate; use effective BVM or an authorized BIAD when it reduces pauses.

Indications

- Pulseless patient age 3 days through 15 years.

Treatment steps

- With an advanced airway, use continuous compressions and ventilate 20-30/min. Avoid excessive ventilation.
- Perform rhythm and pulse checks approximately every 2 minutes; keep each pause under 10 seconds.
- Assess glucose, temperature, and reversible causes during resuscitation. Follow PC-01 for asystole/PEA and PC-07 for VF/pulseless VT.
- Use waveform $EtCO_2$ trends to improve CPR quality and identify possible ROSC; do not use a single $EtCO_2$ value as a termination decision.
- After sustained ROSC, move immediately to PC-08; continue controlled oxygenation/ventilation, frequent BP and glucose reassessment, and early destination notification.

Medications

Naloxone

For suspected opioid-associated arrest, follow UP-04: parenteral naloxone 0.01 mg/kg IV/IO/IM initially; if needed, 0.1 mg/kg, maximum 2 mg per dose and 8 mg cumulative. This is not an intranasal order. Pediatric IN administration requires a product/device-specific Medical Control order pending stocked-product selection. Continue CPR and ventilation without interruption; do not delay resuscitation for naloxone.

Warnings / precautions

- Do not delay CPR, ventilation, or defibrillation for IV access, medication, or endotracheal intubation.
- Naloxone is only for suspected opioid-associated arrest and does not replace high-quality CPR and ventilation.
- Do not pause CPR after a shock to check a pulse; resume immediately until the next planned rhythm check.
- Do not use a single $EtCO_2$ value alone to make termination decisions.

Clinical pearls

- Effective ventilation is especially important because most pediatric arrests are respiratory/asphyxial in origin.
- If pediatric pads/attenuator are unavailable, standard defibrillation equipment should not be withheld or delayed.

Related pathways / contacts

AR-05 Pediatric Airway - Open for pediatric BVM and airway support.

AR-06 Pediatric Failed Airway - Open for ineffective oxygenation/ventilation or unsafe further airway attempts.

PC-01 Pediatric Asystole / PEA - Open for nonshockable pulseless arrest.

PC-07 Pediatric VF / Pulseless VT - Open for shockable pulseless arrest.

PC-08 Pediatric Post-Resuscitation Care - Open immediately after ROSC.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

Revision: 2026-08-14 | Last verified: 2026-08-14